# UMC 0.11um Logic and Mixed-Mode Al Advanced Enhancement Vertical NPN BJT SPICE Model (Gummel-Poon) Document

**Version 0.1 Phase 1**
**2015/01/05**

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# 1 Introduction

## 1.1 Revision History

**Table 1-1 Revision history**

| Version | Date | Author | Remark |
|---|---|---|---|
| v0.1_p1 | 2015/01/05 | Wei Cheng Lin | 1. Original Release. |

## 1.2 General Information

This document serves as a reference to the design works of products that will be manufactured by UMC using the following process.

UMC 0.11 um Logic and Mixed-Mode Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

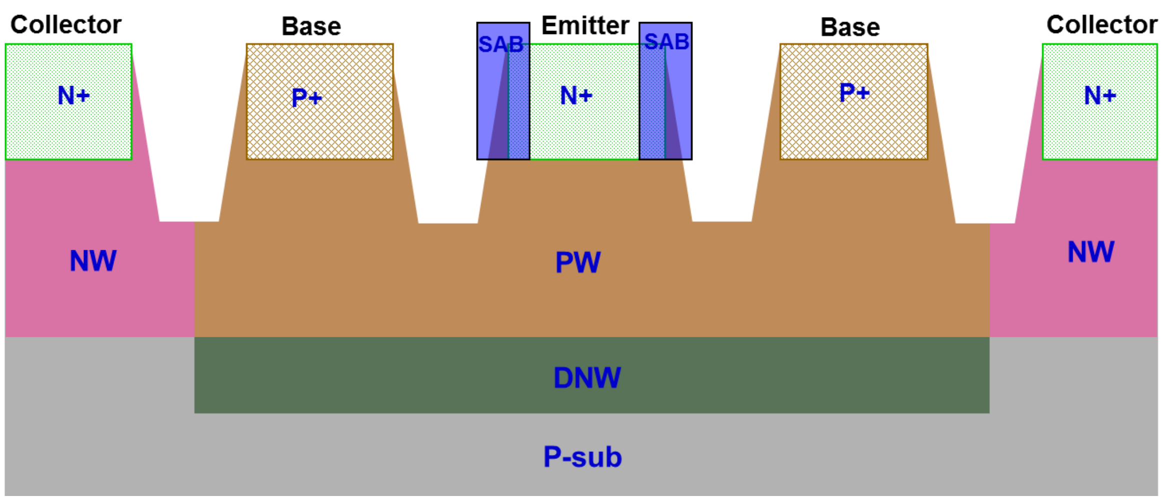


**Figure 1-1 BJT Layout of NPN**

## 1.3 Model Usage

The compact model is provided in various formats. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of simulator might be different.

Synopsys HSPICE release 2013.3SP2
Cadence SPECTRE version 13.1.0.066
Mentor Graphics ELDO version 6.11_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below.

TT: Typical BJT model
FF: Fast BJT model (Maximum Beta model)
SS: Slow BJT model (Minimum Beta model)

Follow these steps to invoke the model file during simulation.

**For Synopsys HSPICE**
- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator.
.lib "./ModelFileName.lib" CaseName
, where CaseName could be TT, FF, SS.

**For Cadence Spectre**
- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator.
include "./ModelFileName.lib.scs" section=CaseName
, where CaseName could be TT, FF, or SS.

**For Mentor Graphics Eldo**
- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.
- Add the following statement in the input file to the simulator.
.lib "./ModelFileName.lib.eldo" CaseName
, where CaseName could be TT, FF, SS.

As stated above that BJT model is provided and this model is valid as the following geometry, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

**Geometry description:**

**Table 1-2 Vertical BJT device**

| Vertical NPN | npn_sv45x45_ L110AE | npn_sv90x90_ L110AE | npn_sv135x135_ L110AE |
|---|---|---|---|
| Emitter Area | 4.5X4.5 $\mu m^2$ | 9X9 $\mu m^2$ | 13.5X13.5 $\mu m^2$ |
| Base Area | 11.7X11.7 $\mu m^2$ | 16.2X16.2 $\mu m^2$ | 20.7X20.7 $\mu m^2$ |
| SAB (Salicide Block) overlap diffusion | Yes | Yes | Yes |

**Temperature range:**

-40 ℃ ~ +150 ℃

# 2 NPN BJT Model Verification

## 2.1 Silicon Wafer For Modeling

The information of the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

**Table 2-1 Test wafer information for BJT**

| | BJT NPN model |
|---|---|
| Mask ID | BSW0QA-A0 |
| Lot ID | D58YR.1 |
| Wafer Number | 03 |

## 2.2 Measurement Conditions

The model is verified to measurement data under following bias condition

**Gummel Plot:**

**Table 2-2 Gummel Plot measurement condition of NPN**

| Gummel Plot | | | |
|---|---|---|---|
| | Start | Stop | Step |
| $V_{BE}$(V) | 0.1 | 1.0 | 0.01 |
| $V_{CB}$(V) | 1 | 1 | 1 |

**Output characteristics:**

**Table 2-3 Output characteristic measurement conditions of NPN**

| Output Characteristics | | | |
|---|---|---|---|
| | Start | Stop | Step |
| $V_{CE}$(V) | 0 | 3.3 | 0.033 |
| $I_B$(μA) | 1 | 5 | 1 |

*Gummel-Poon* model has its essential inaccuracy in reverse region, because this model does not have a parameter for the reverse saturation current. Hence, its uses the forward saturation current for both forward and reverse region. In addition, the base resistance parameters ($R_B$, $R_{BM}$) were extracted from the forward base current. That is the reason that the reverse base current is not modeled well at high current level.

## 2.3 NPN_SV45X45_L110AE Model Verification

### 2.3.1 NPN_SV45X45_L110AE model verification at 25 °C

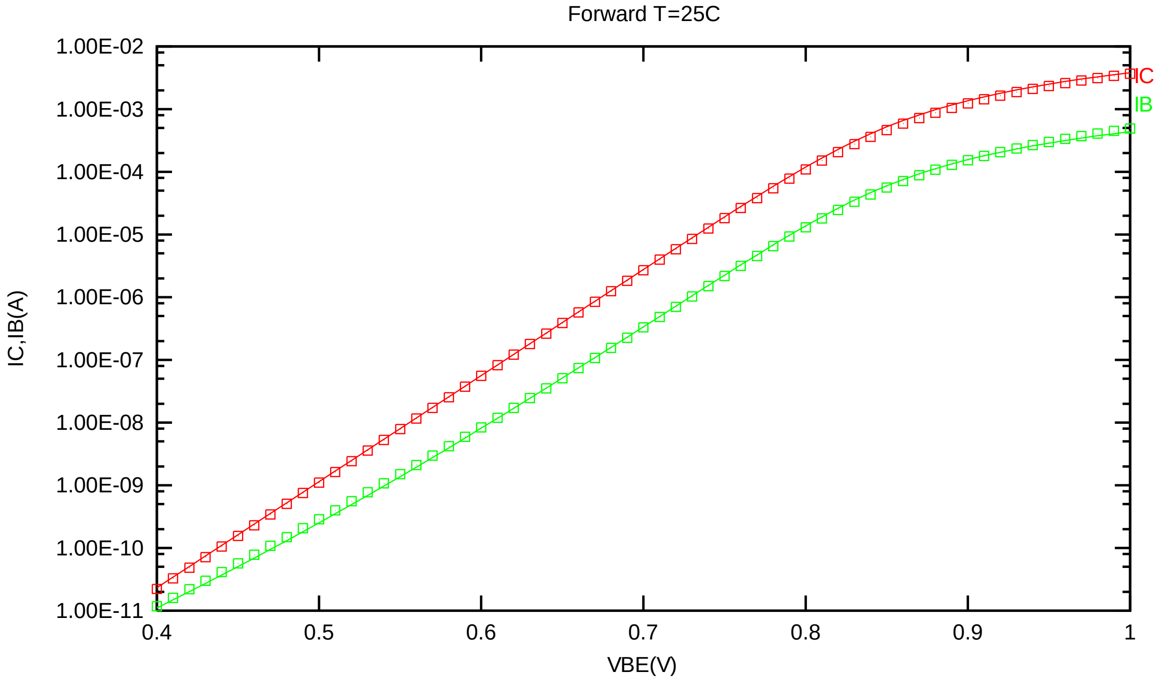


**Figure 2-1 Forward gummel plot of NPN_SV45X45_L110AE at 25 °C, VCB = 1V**

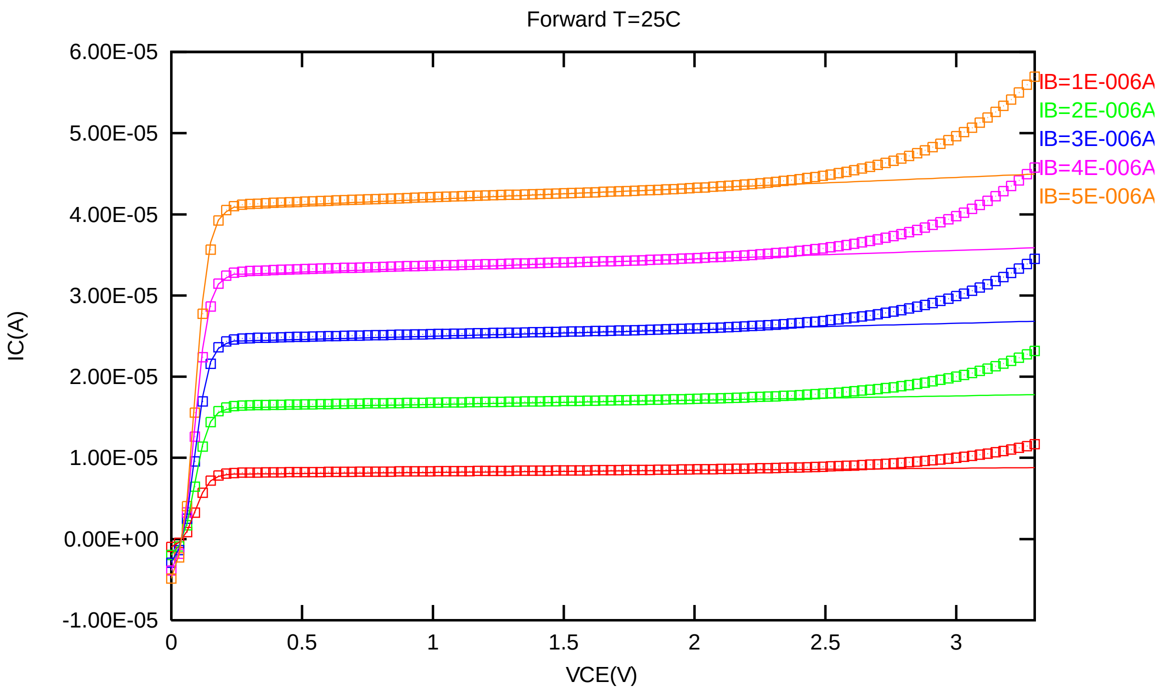


**Figure 2-2 Forward output characteristics of NPN_SV45X45_L110AE at 25 °C**

### 2.3.2 NPN_SV45X45_L110AE model verification at 85 °C

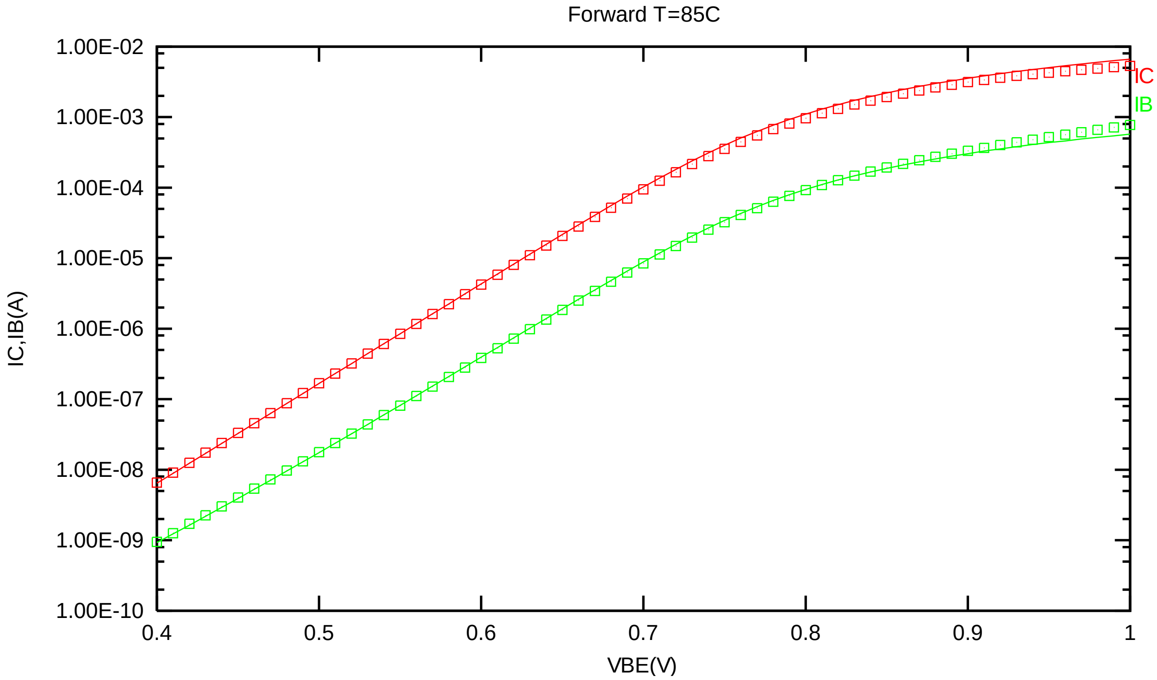


**Figure 2-3 Forward gummel plot of NPN_SV45X45_L110AE at 85 °C, VCB = 1V**

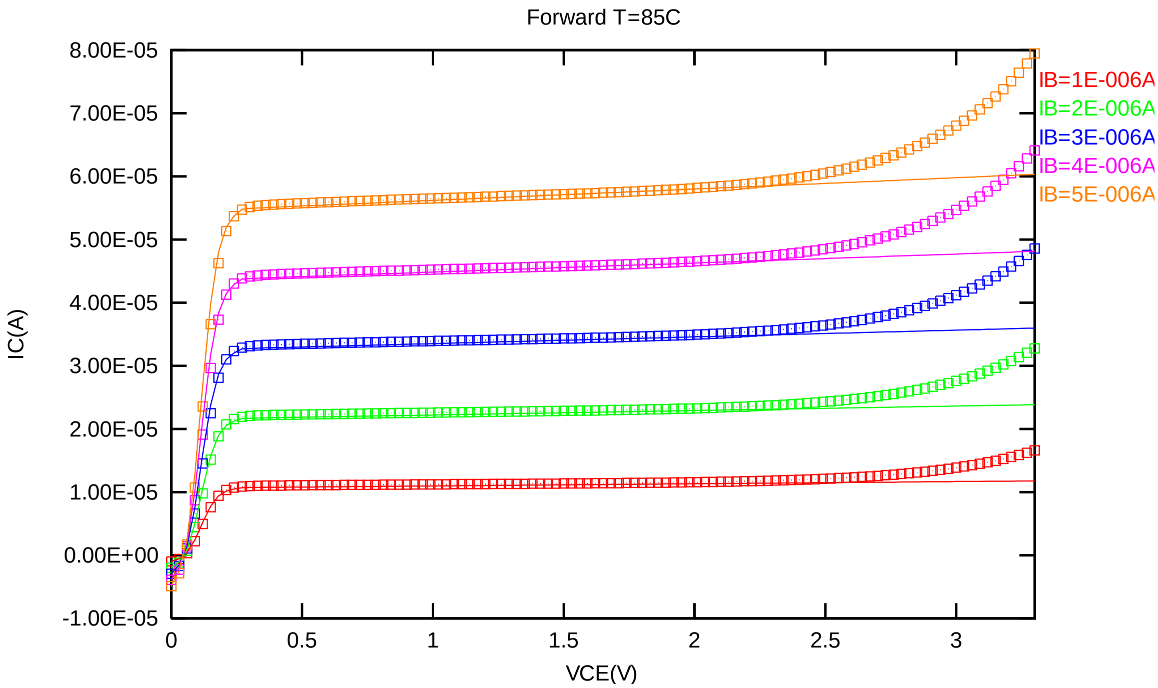


**Figure 2-4 Forward output characteristics of NPN_SV45X45_L110AE at 85 °C**

### 2.3.3 NPN_SV45X45_L110AE model verification at 150 °C

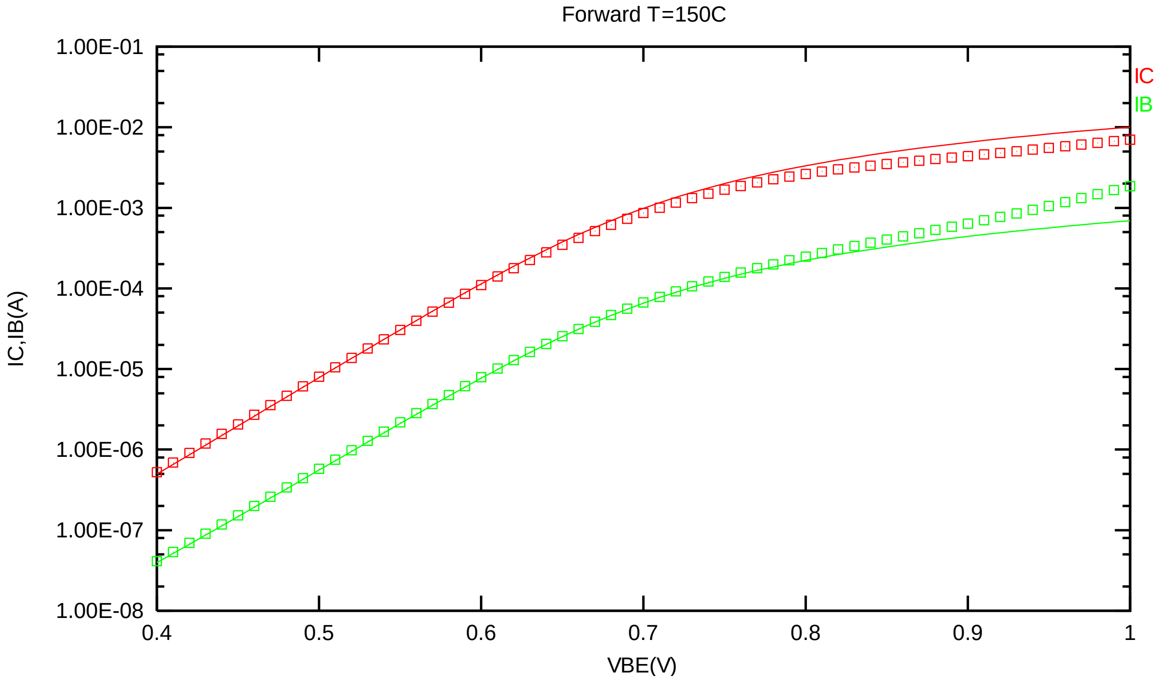


**Figure 2-5 Forward gummel plot of NPN_SV45X45_L110AE at 150 °C, VCB = 1V**

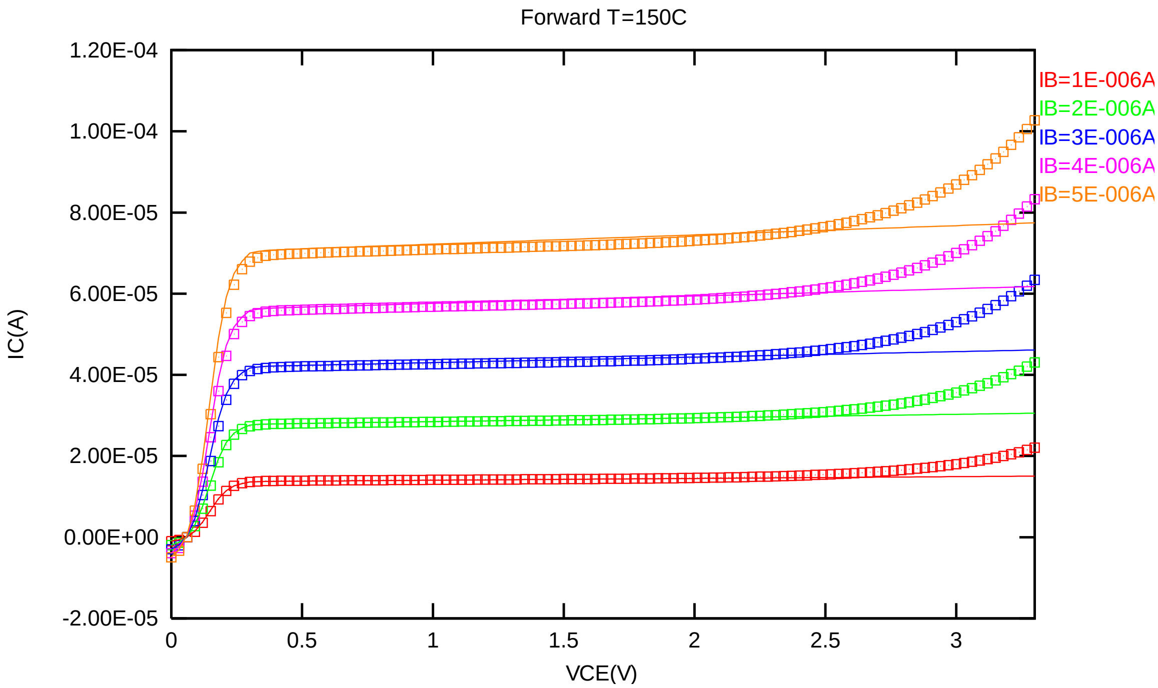


**Figure 2-6 Forward output characteristics of NPN_SV45X45_L110AE at 150 °C**

### 2.3.4 NPN_SV45X45_L110AE model verification at 0 °C

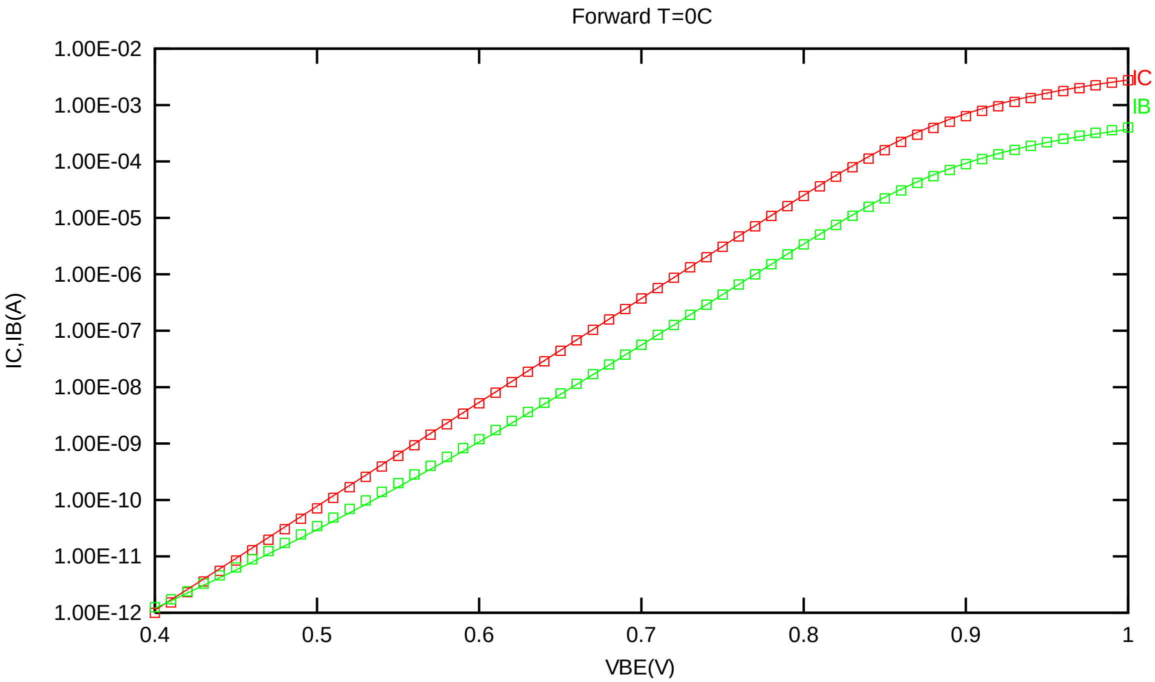


**Figure 2-7 Forward gummel plot of NPN_SV45X45_L110AE at 0 °C, VCB =1V**

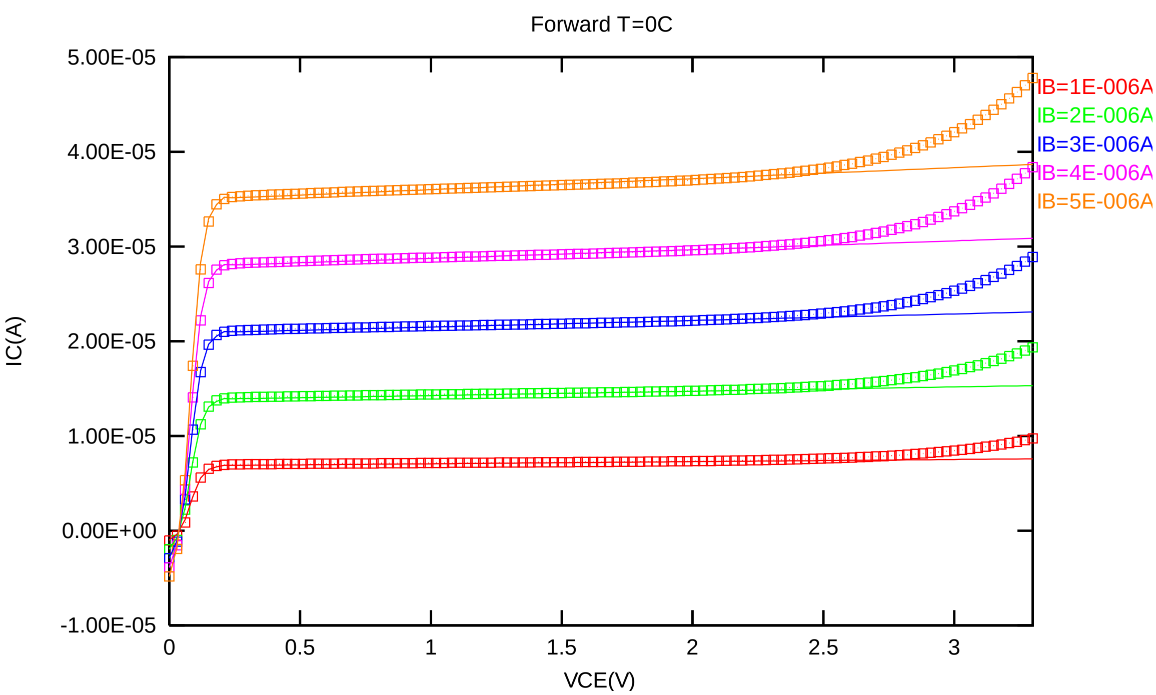


**Figure 2-8 Forward output characteristics of NPN_SV45X45_L110AE at 0 °C**

### 2.3.5 NPN_SV45X45_L110AE model verification at -40 °C

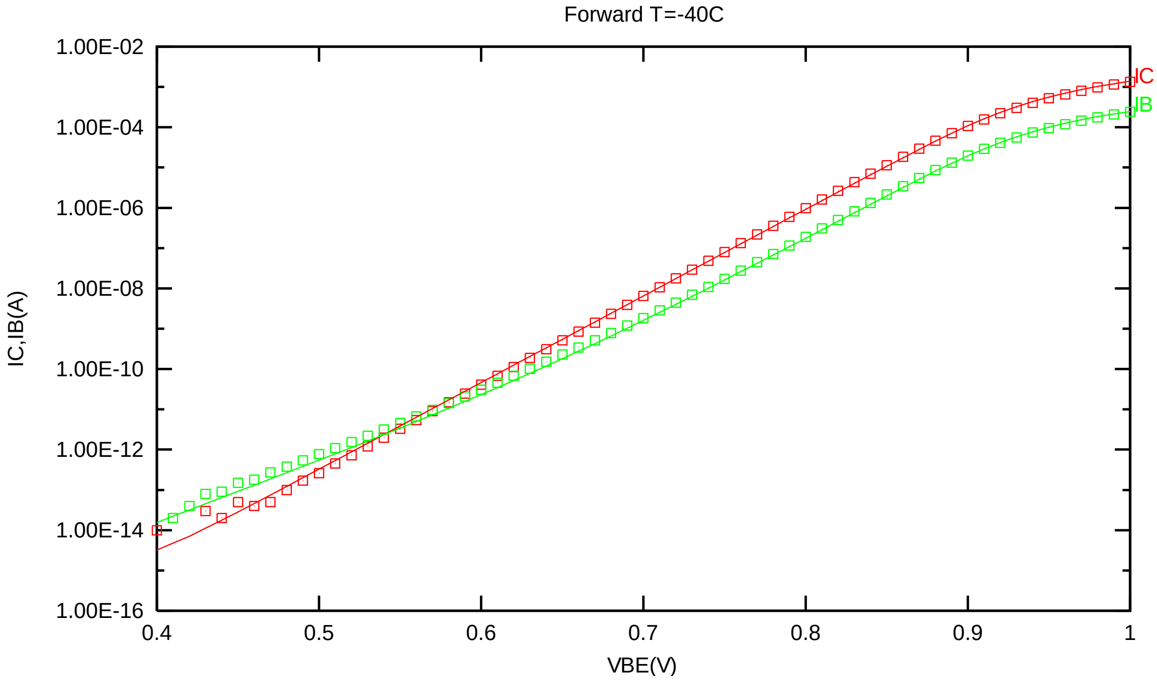


**Figure 2-9 Forward gummel plot of NPN_SV45X45_L110AE at -40 °C, VCB = 1V**

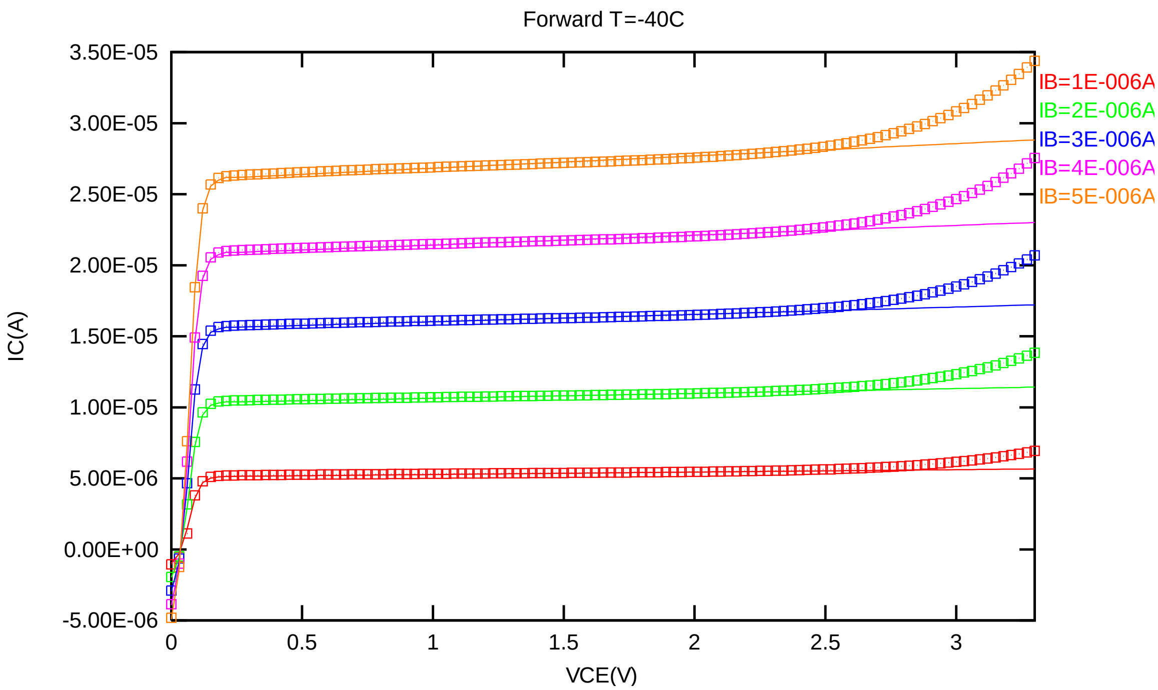


**Figure 2-10 Forward output characteristics of NPN_SV45X45_L110AE at -40 °C**

### 2.3.6 NPN_SV45X45_L110AE NPN model temperature effect

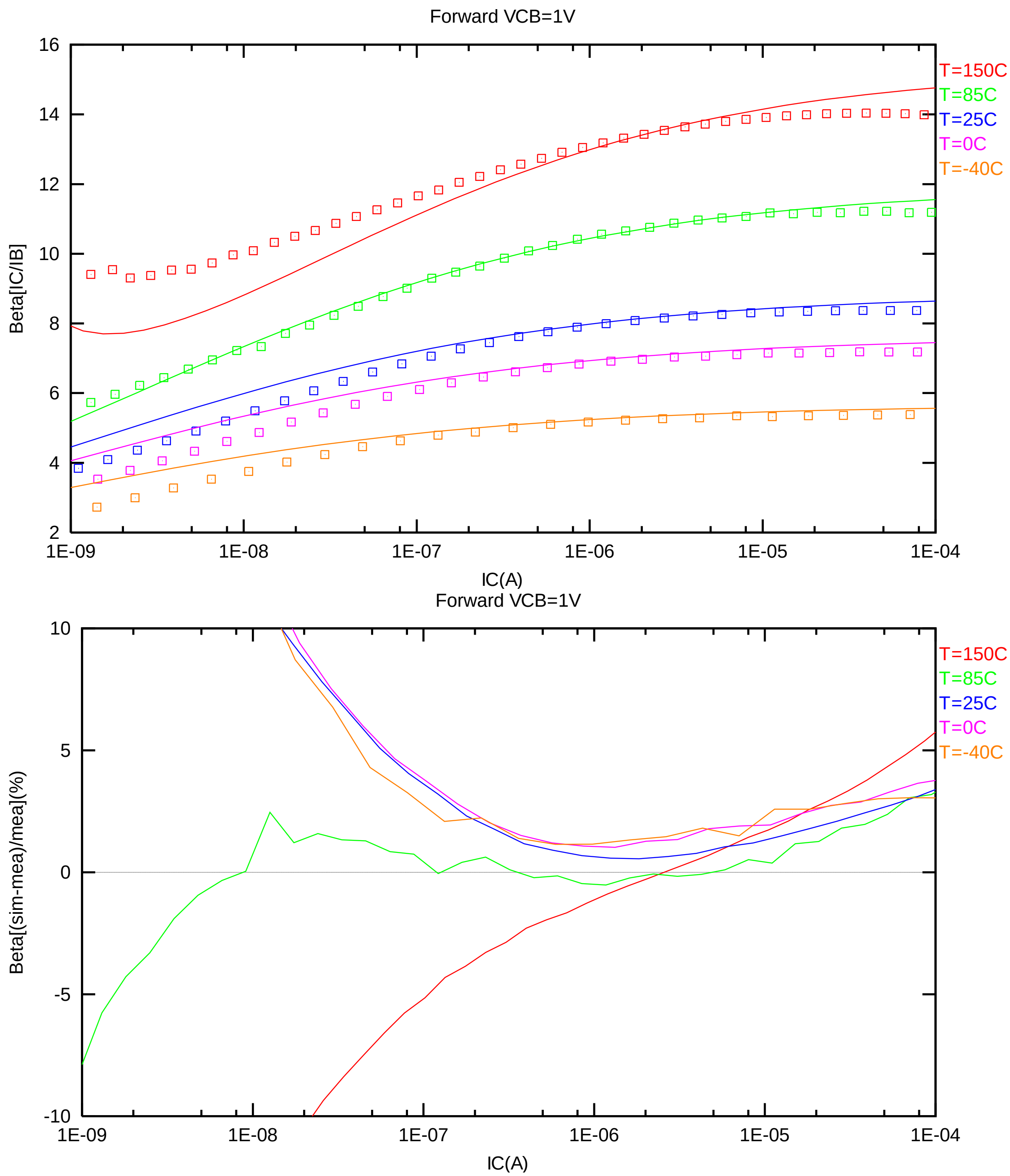


**Figure 2-11 Current gain vs. IC at different temperatures of NPN_SV45X45_L110AE, VCB = 1V**

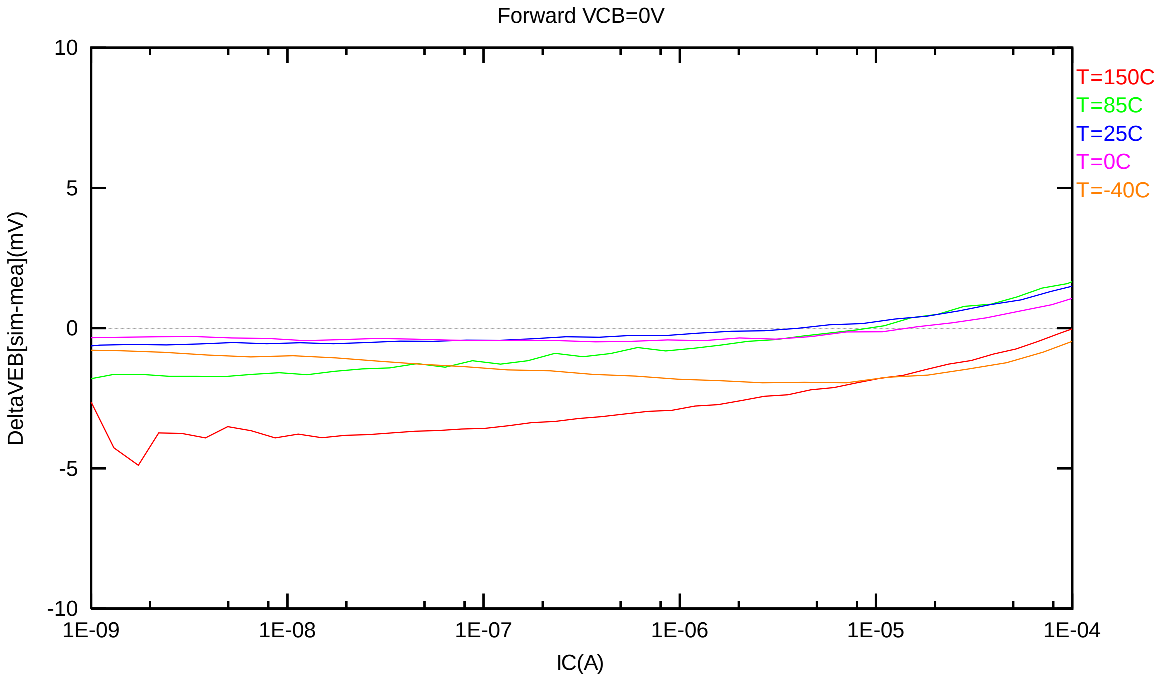


**Figure 2-12 DeltaVEB vs. IC at different temperatures of NPN_SV45X45_L110AE, VCB = 1V**

## 2.4 NPN_SV90X90_L110AE Model Verification

### 2.4.1 NPN_SV90X90_L110AE model verification at 25 °C

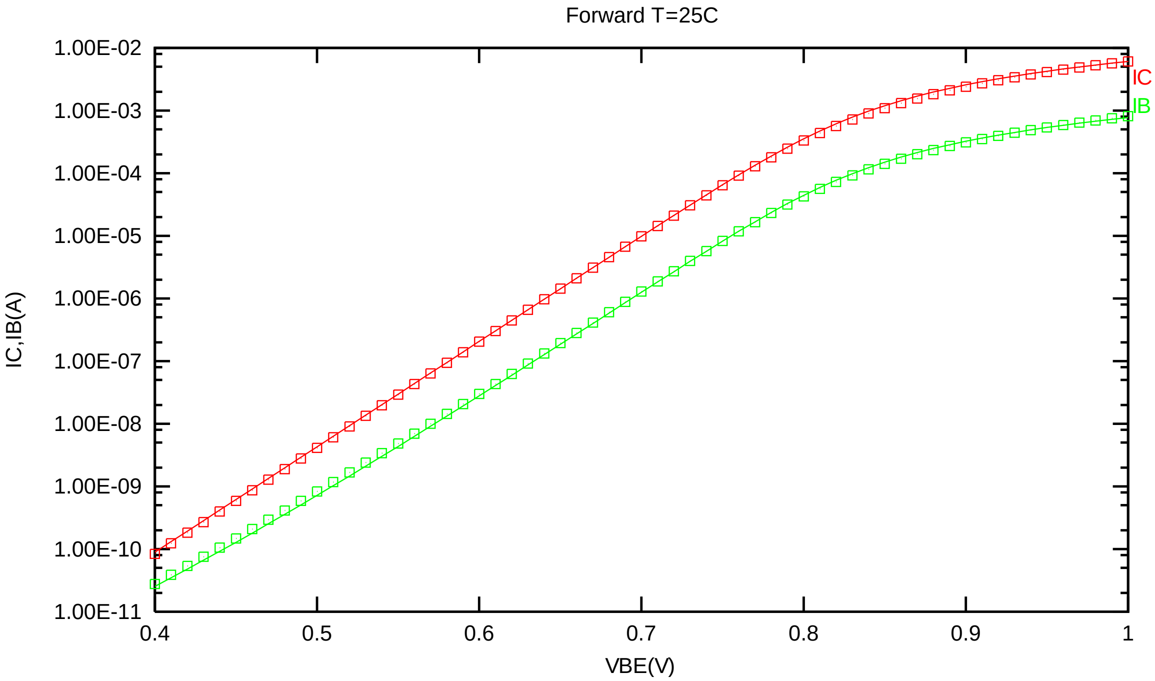


**Figure 2-13 Forward gummel plot of NPN_SV90X90_L110AE at 25 °C, VCB = 1V**

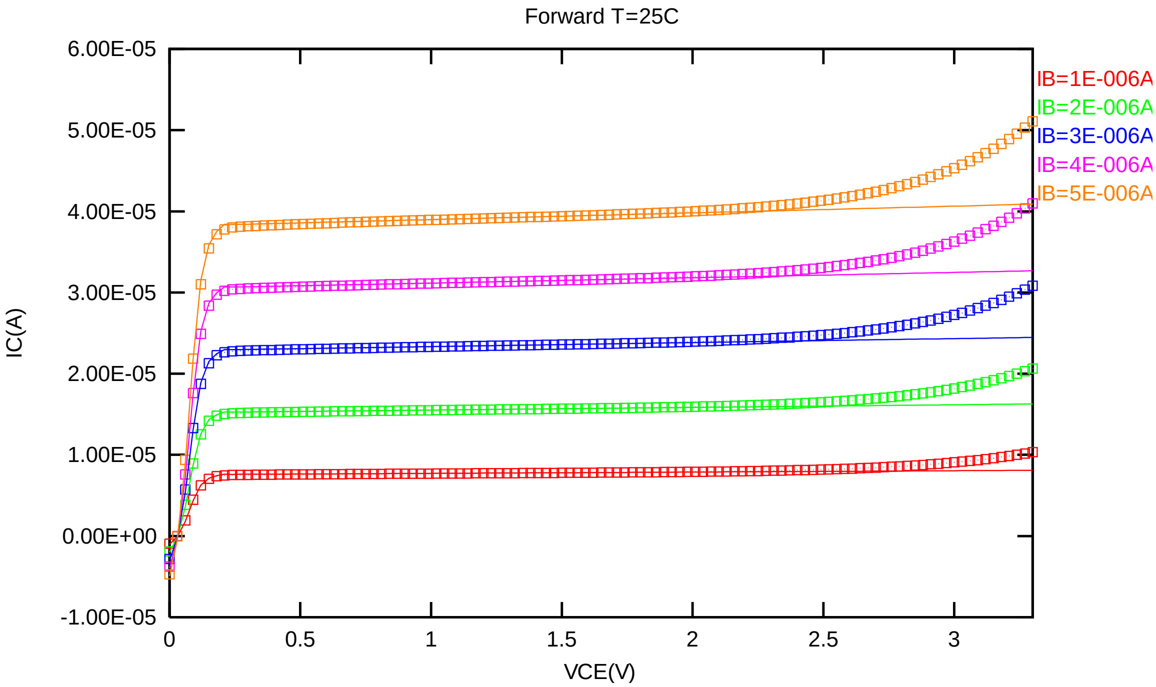


**Figure 2-14 Forward output characteristics of NPN_SV90X90_L110AE at 25 °C**

### 2.4.2 NPN_SV90X90_L110AE model verification at 85 °C

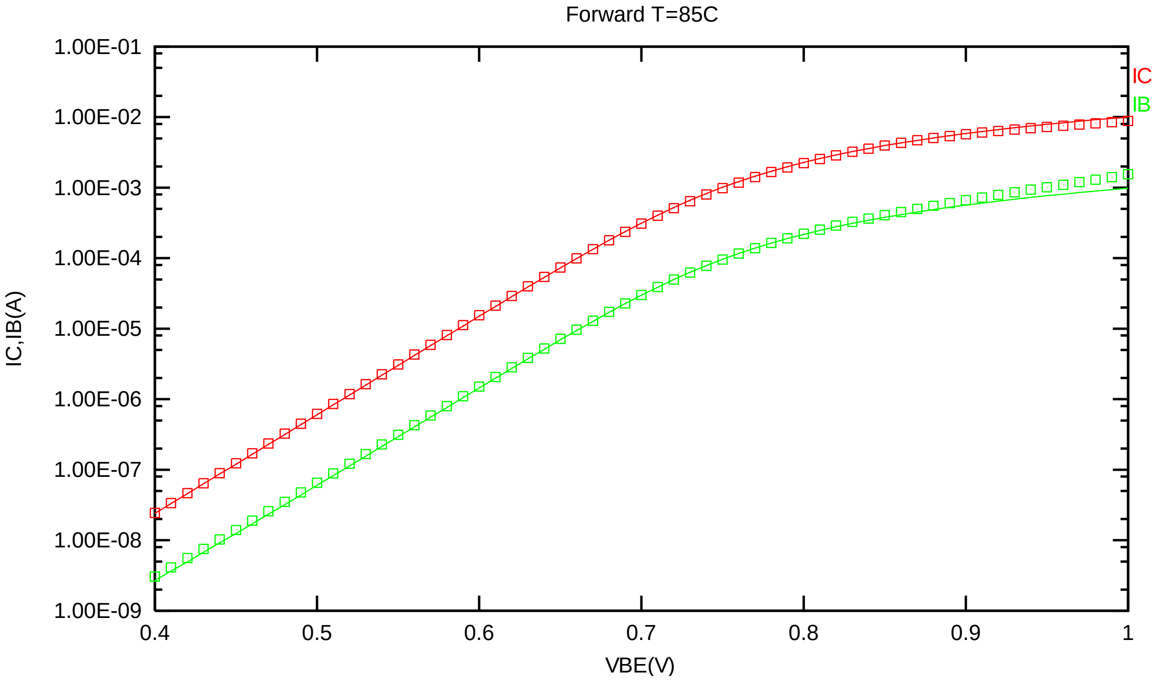


**Figure 2-15 Forward gummel plot of NPN_SV90X90_L110AE at 85 °C, VCB = 1V**

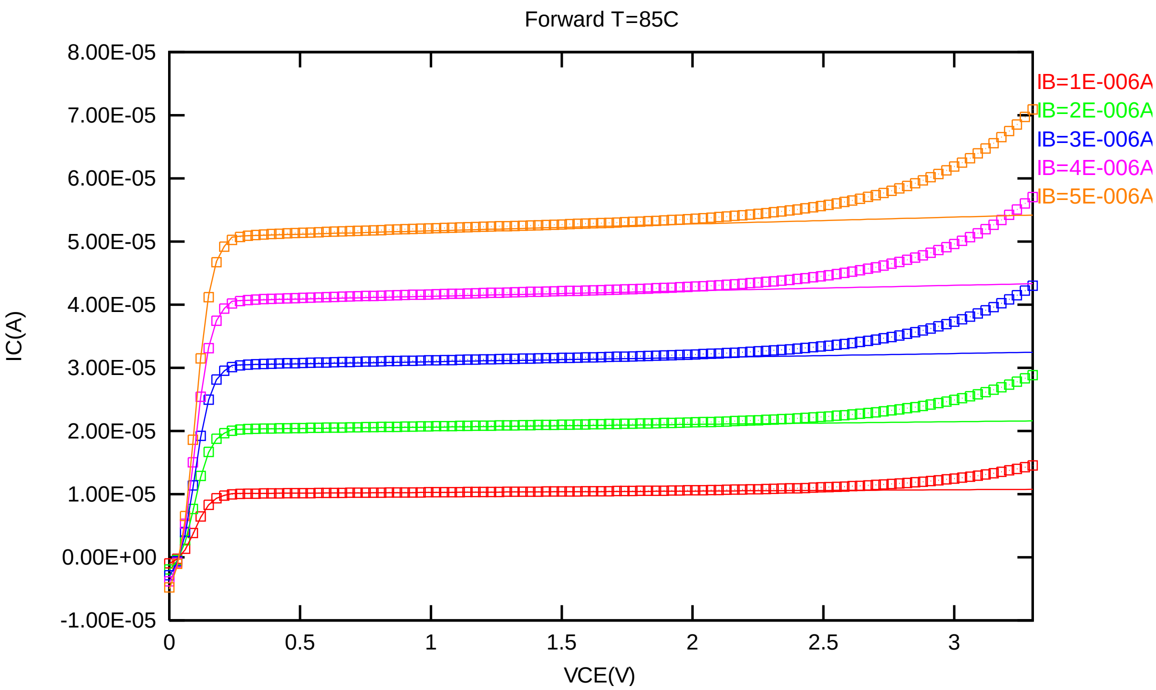


**Figure 2-16 Forward output characteristics of NPN_SV90X90_L110AE at 85 °C**

### 2.4.3 NPN_SV90X90_L110AE model verification at 150 °C

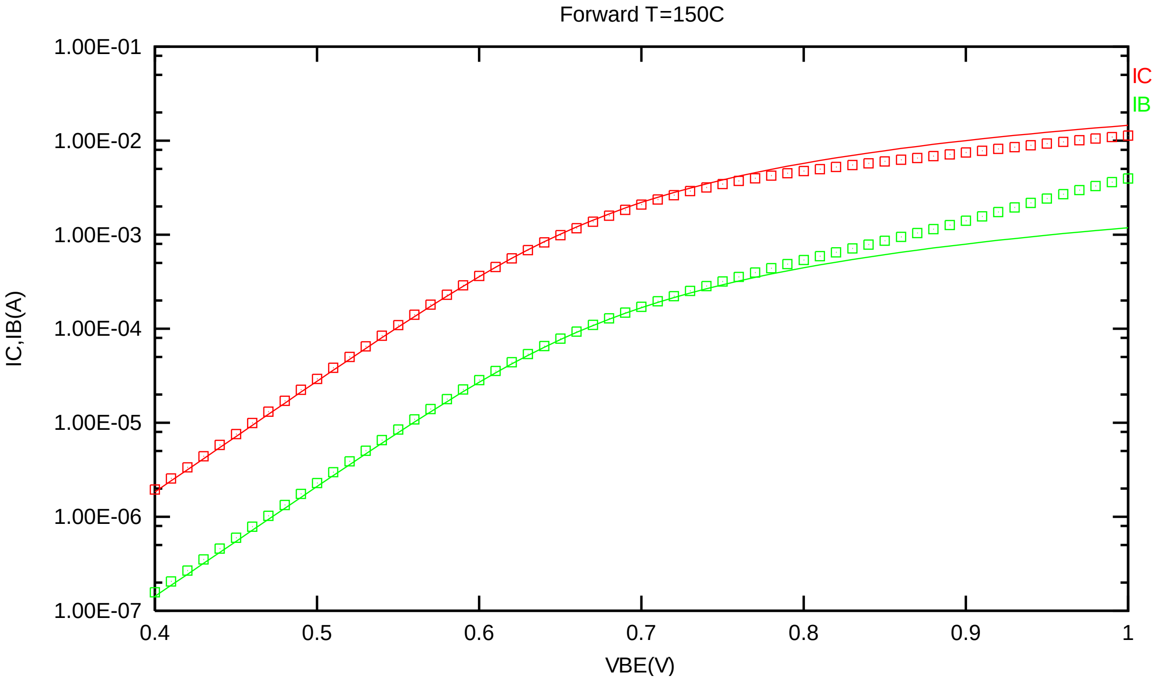


**Figure 2-17 Forward gummel plot of NPN_SV90X90_L110AE at 150 °C, VCB = 1V**

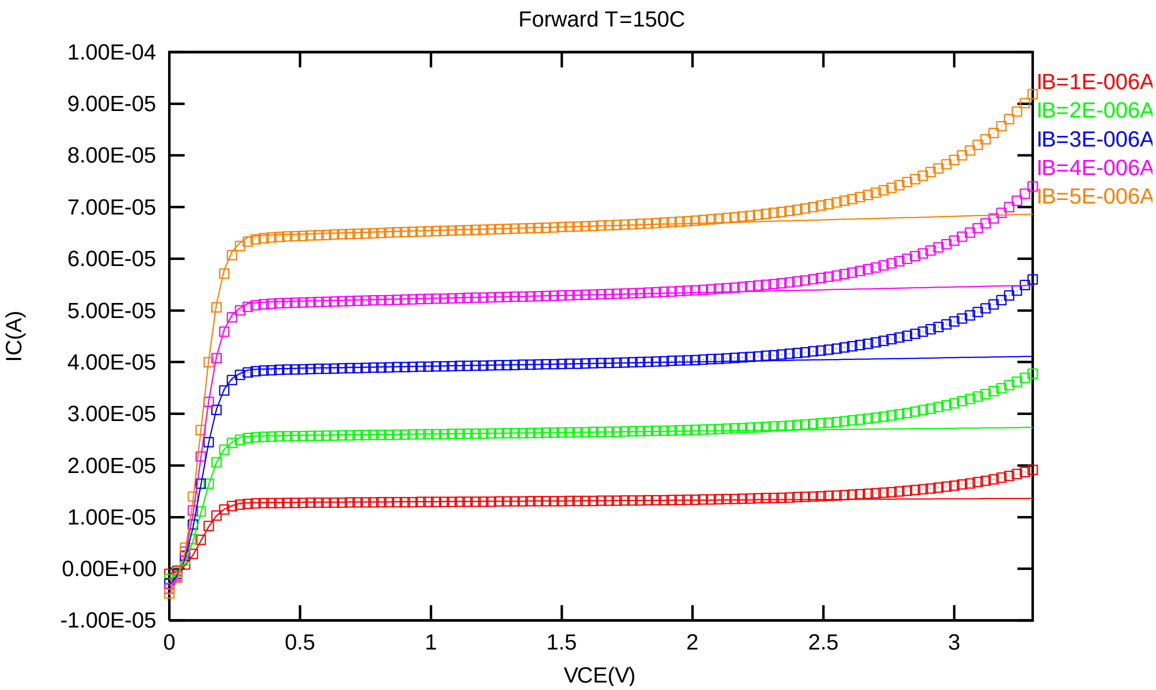


**Figure 2-18 Forward output characteristics of NPN_SV90X90_L110AE at 150 °C**

### 2.4.4 NPN_SV90X90_L110AE model verification at 0 ℃

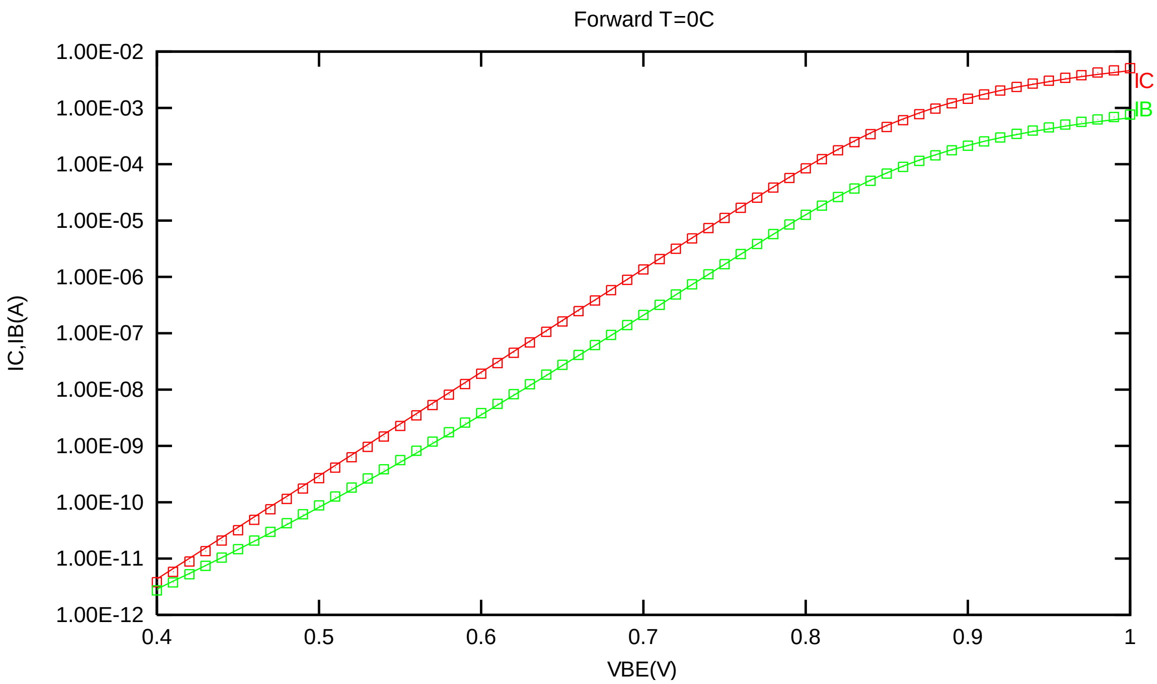


**Figure 2-19 Forward gummel plot of NPN_SV90X90_L110AE at 0 ℃, VCB = 1V**

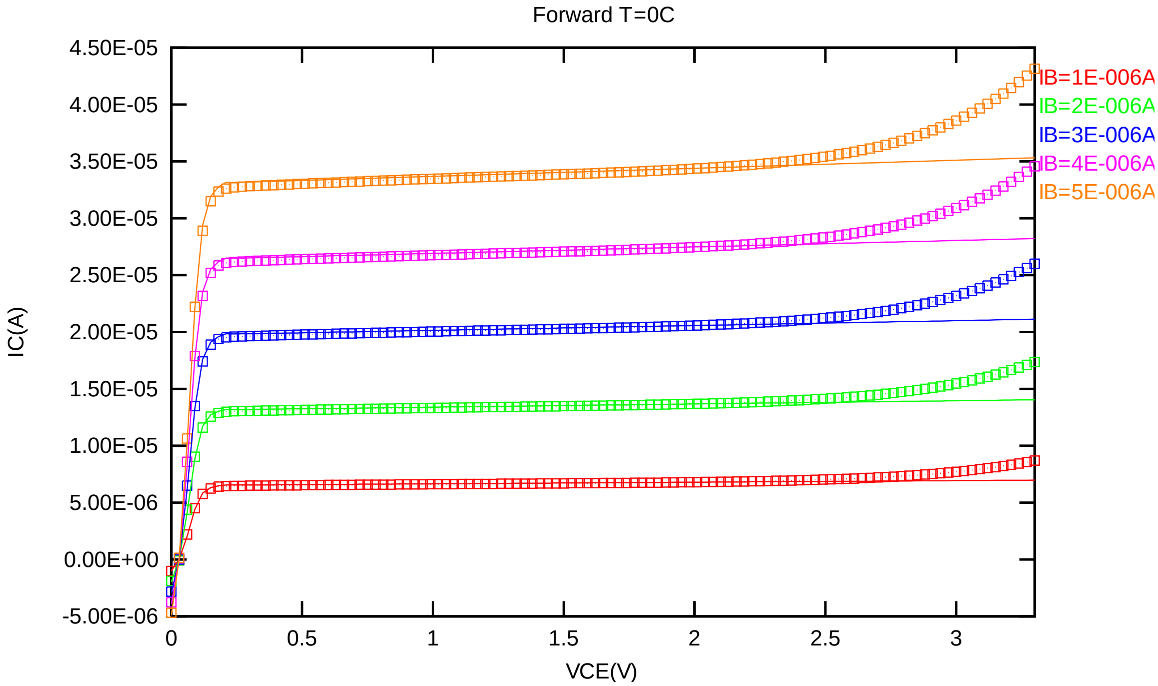


**Figure 2-20 Forward output characteristics of NPN_SV90X90_L110AE at 0 ℃**

### 2.4.5 NPN_SV90X90_L110AE model verification at -40 °C

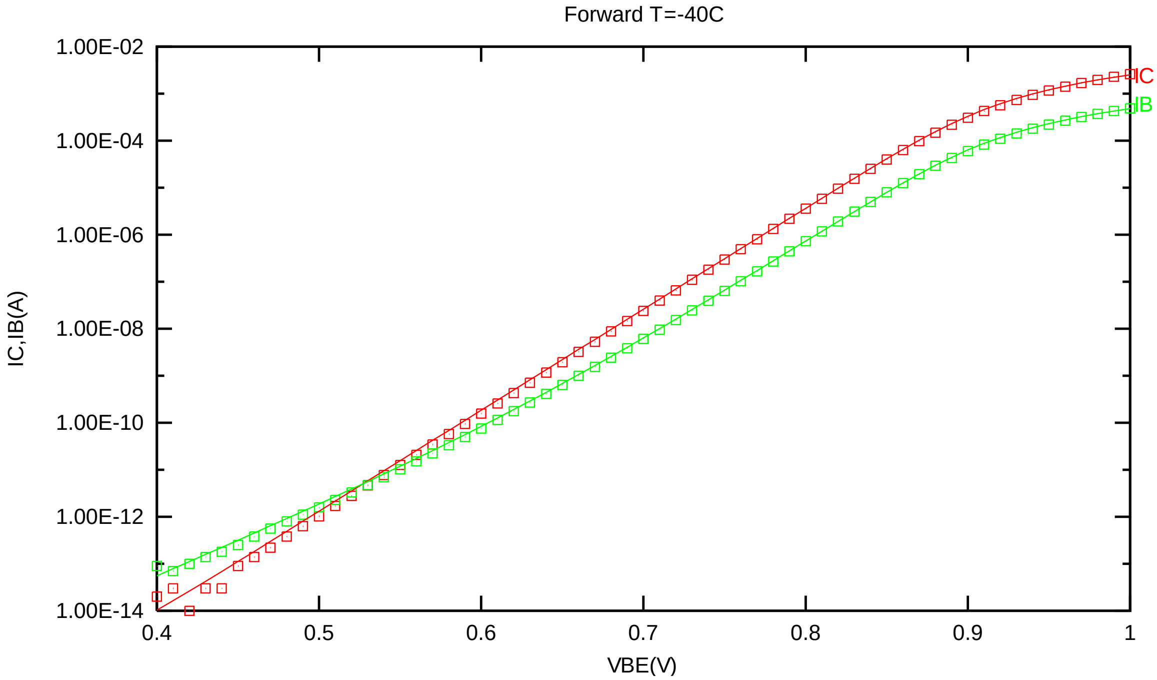


**Figure 2-21 Forward gummel plot of NPN_SV90X90_L110AE at -40 °C, VCB =1V**

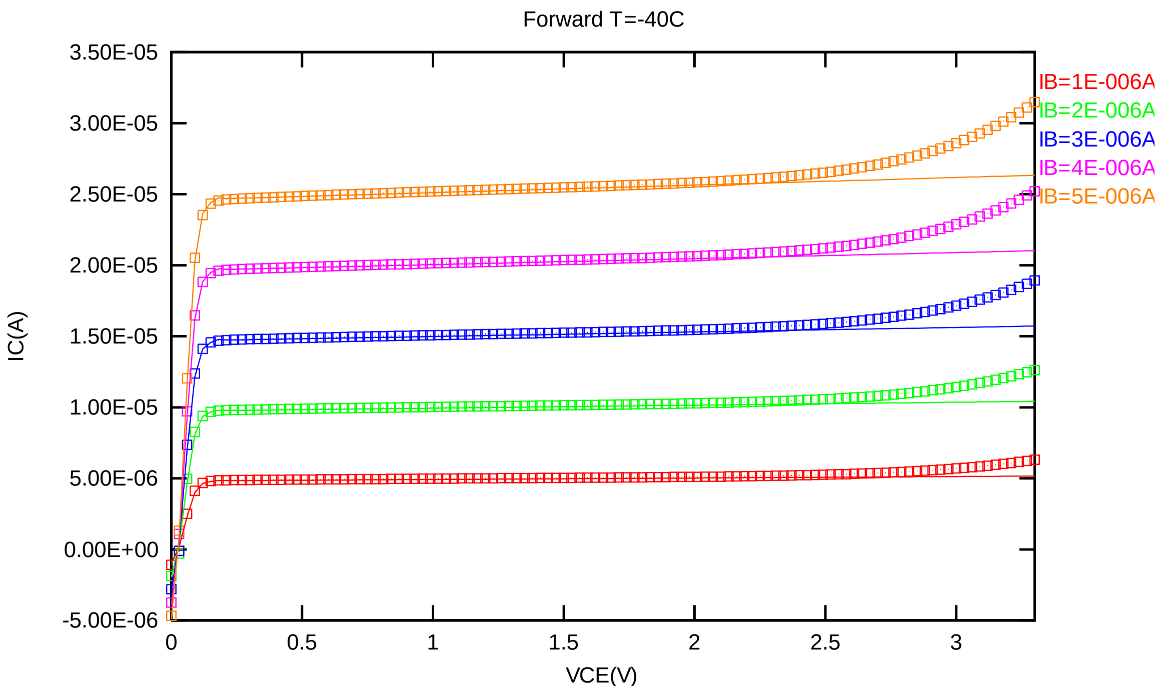


**Figure 2-22 Forward output characteristics of NPN_SV90X90_L110AE at -40 °C**

### 2.4.6 NPN_SV90X90_L110AE NPN model temperature effect

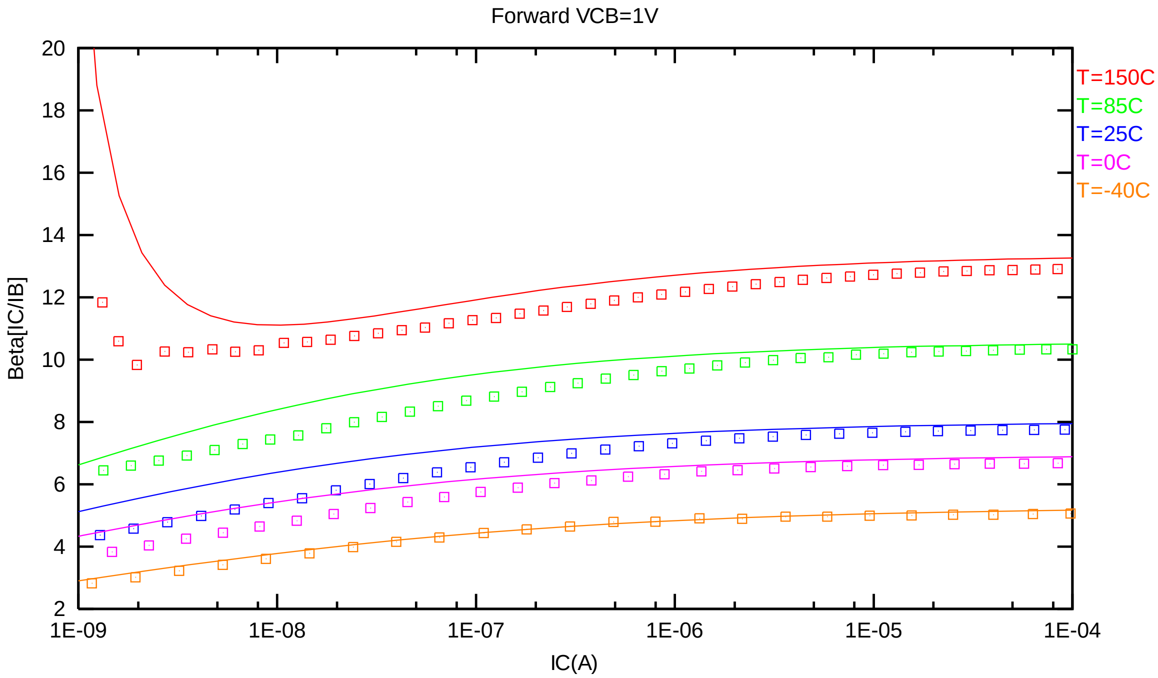


**Figure 2-23 Current gain vs. IC at different temperatures of NPN_SV90X90_L110AE, VCB = 1V**

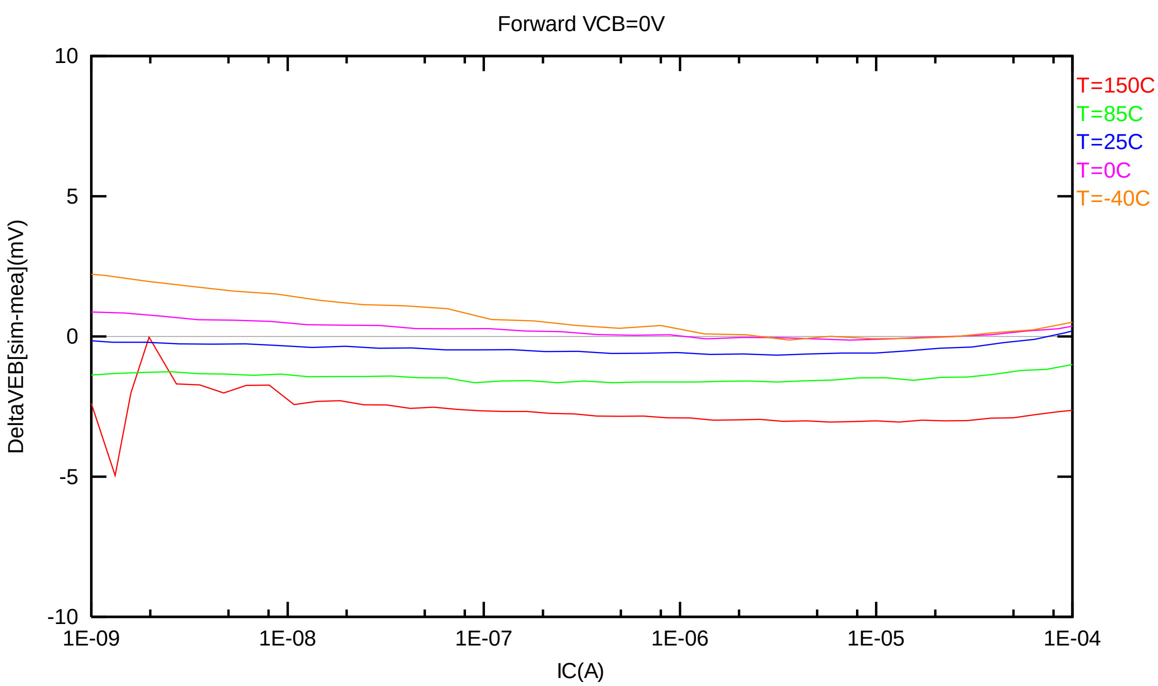


**Figure 2-24 DeltaVEB vs. IC at different temperatures of NPN_SV90X90_L110AE, VCB = 1V**

# 2.5 NPN_SV135X135_L110AE Model Verification

## 2.5.1 NPN_SV135X135_L110AE model verification at 25 °C

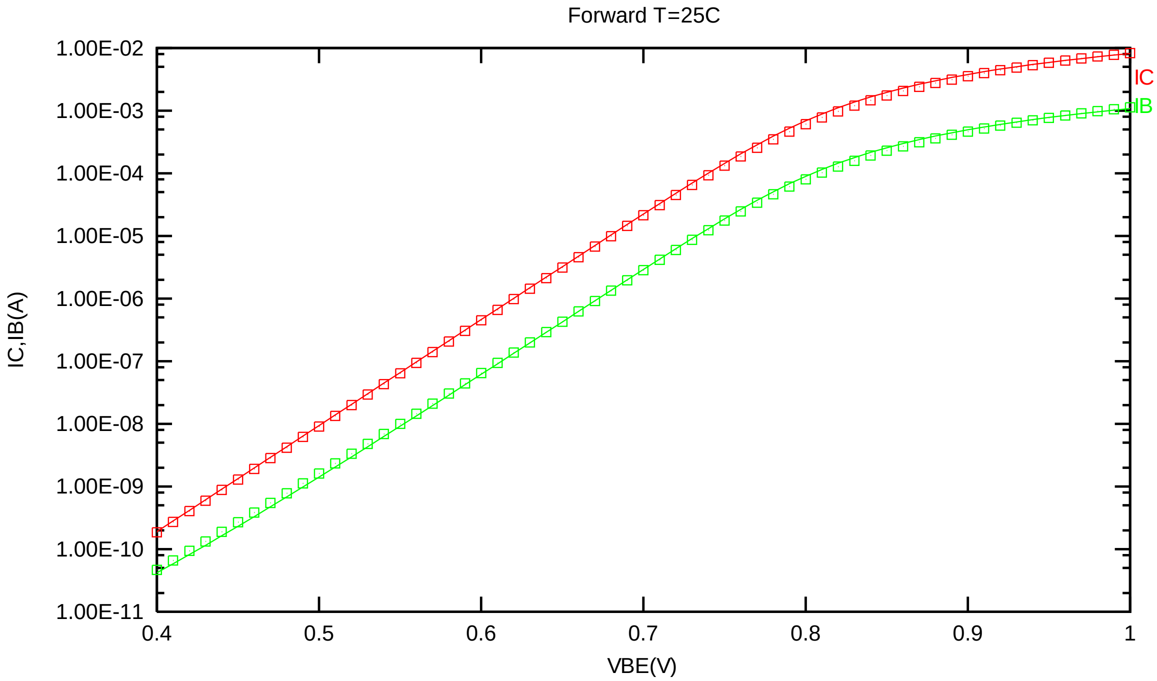


**Figure 2-25 Forward gummel plot of NPN_SV135X135_L110AE at 25 °C, VCB = 1V**

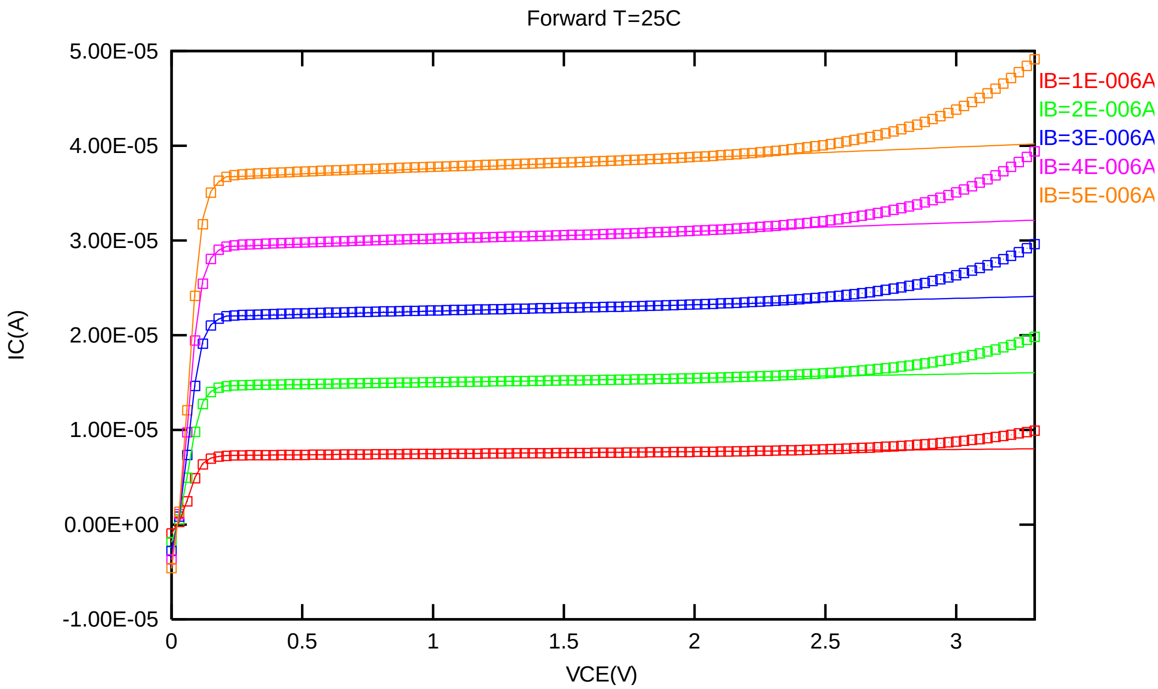

**Figure 2-26 Forward output characteristics of NPN_SV135X135_L110AE at 25 °C**

### 2.5.2 NPN_SV135X135_L110AE model verification at 85 °C

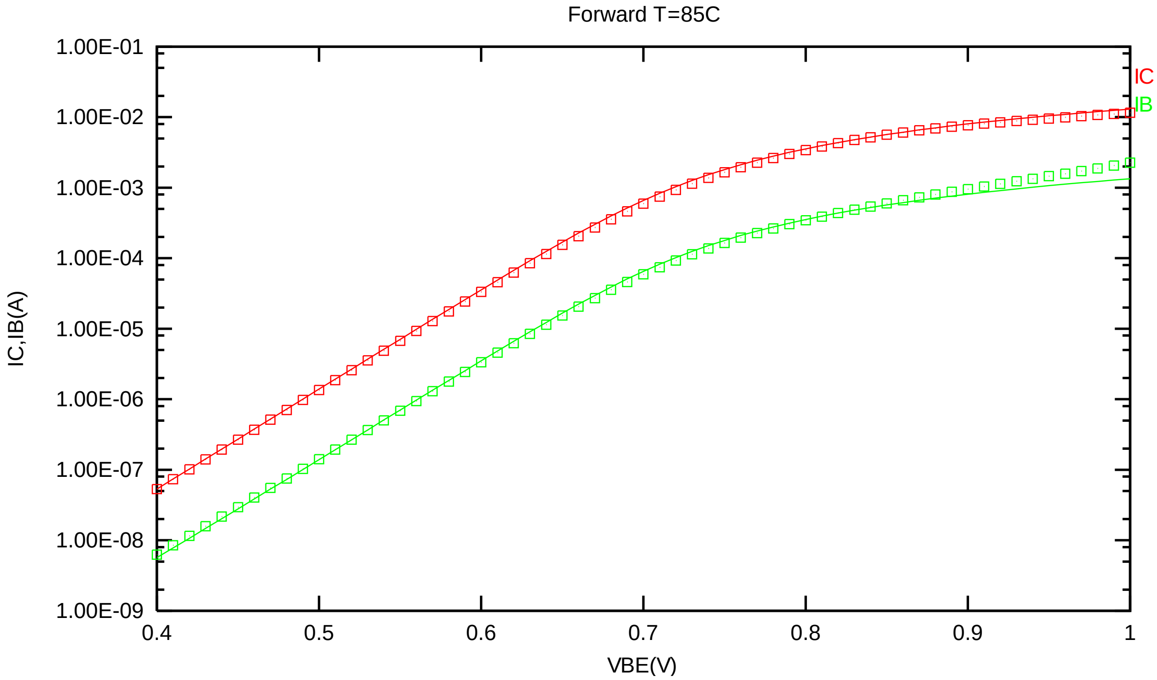


**Figure 2-27 Forward gummel plot of NPN_SV135X135_L110AE at 85 °C, VCB = 1V**

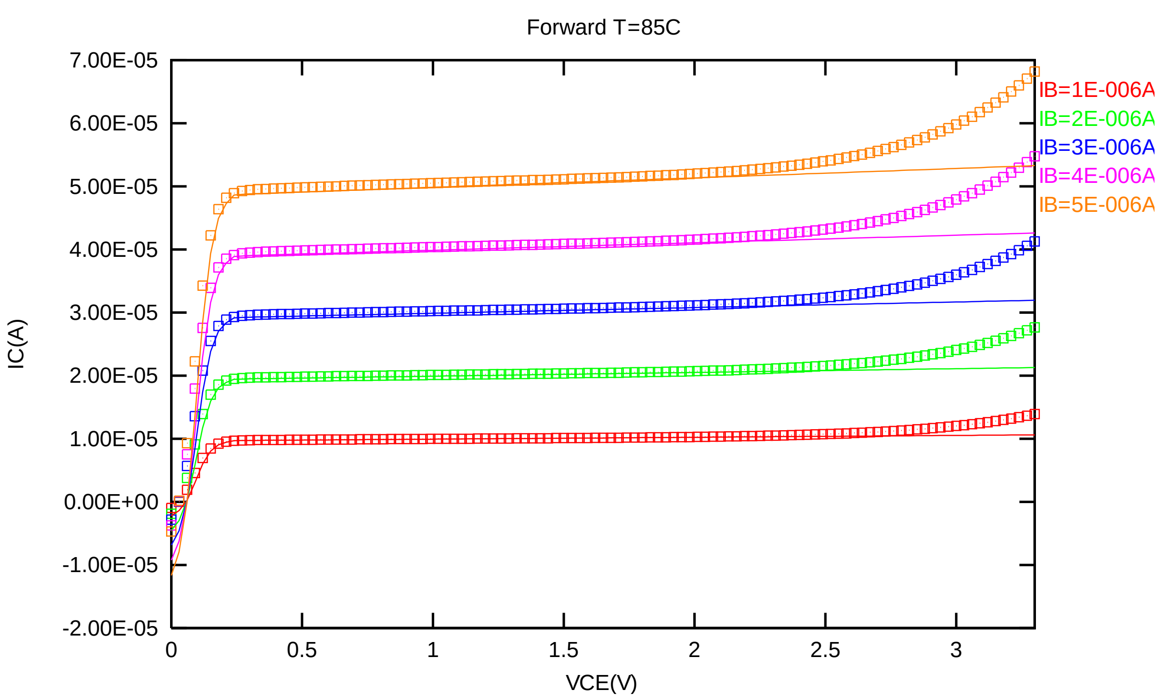

**Figure 2-28 Forward output characteristics of NPN_SV135X135_L110AE at 85 °C**

### 2.5.3 NPN_SV135X135_L110AE model verification at 150 °C

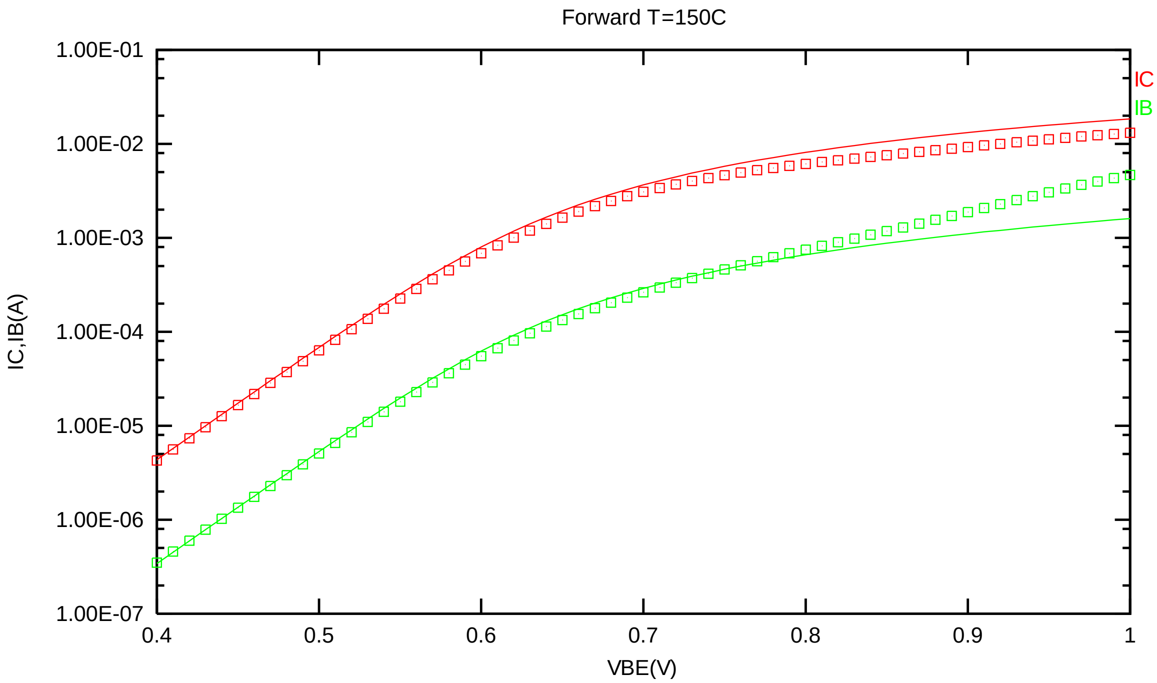


**Figure 2-29 Forward gummel plot of NPN_SV135X135_L110AE at 150 °C, VCB = 1V**

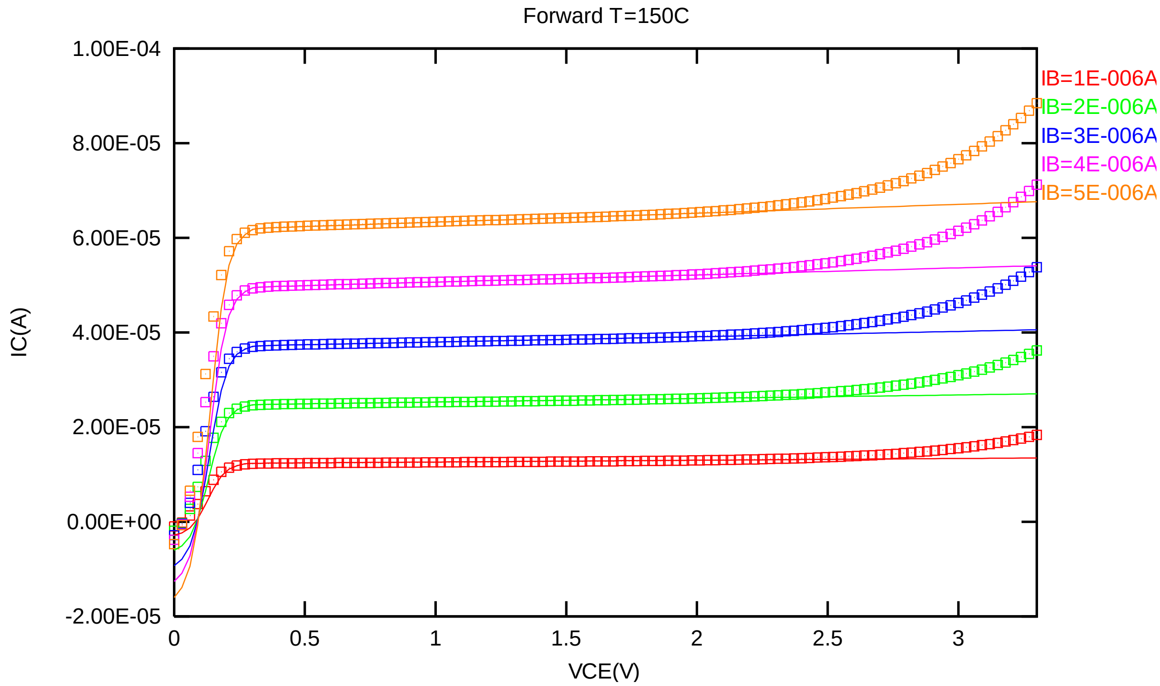


**Figure 2-30 Forward output characteristics of NPN_SV135X135_L110AE at 150 °C**

## 2.5.4 NPN_SV135X135_L110AE model verification at 0 °C

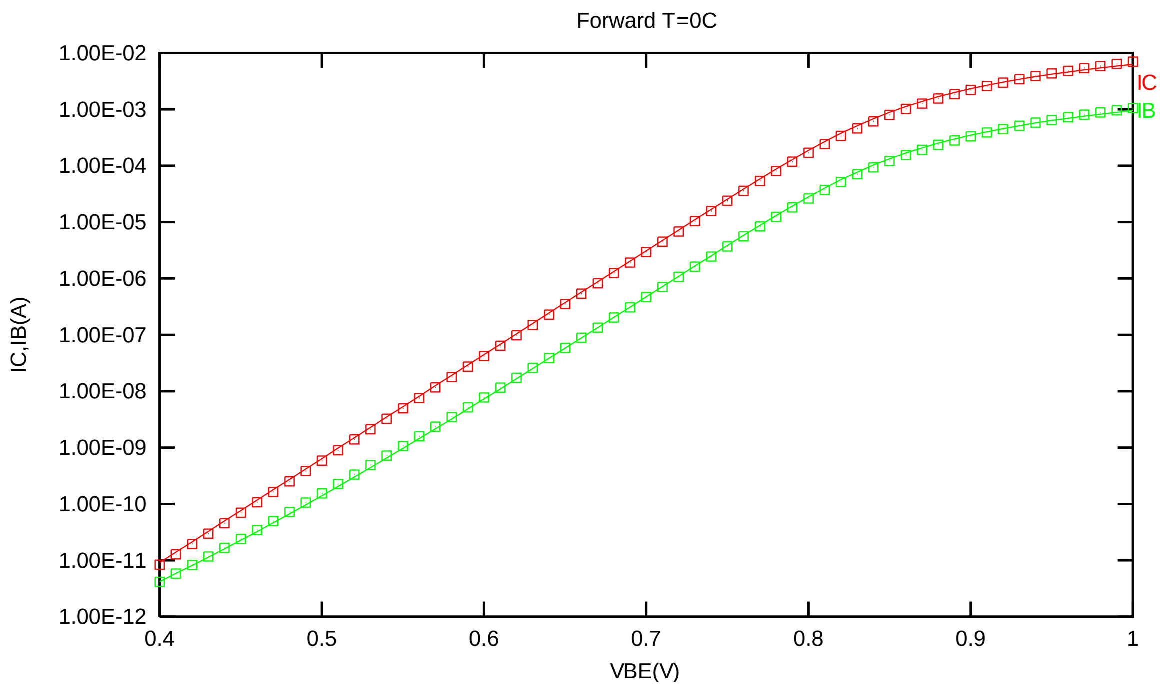


**Figure 2-31 Forward gummel plot of NPN_SV135X135_L110AE at 0 °C, VCB = 1V**

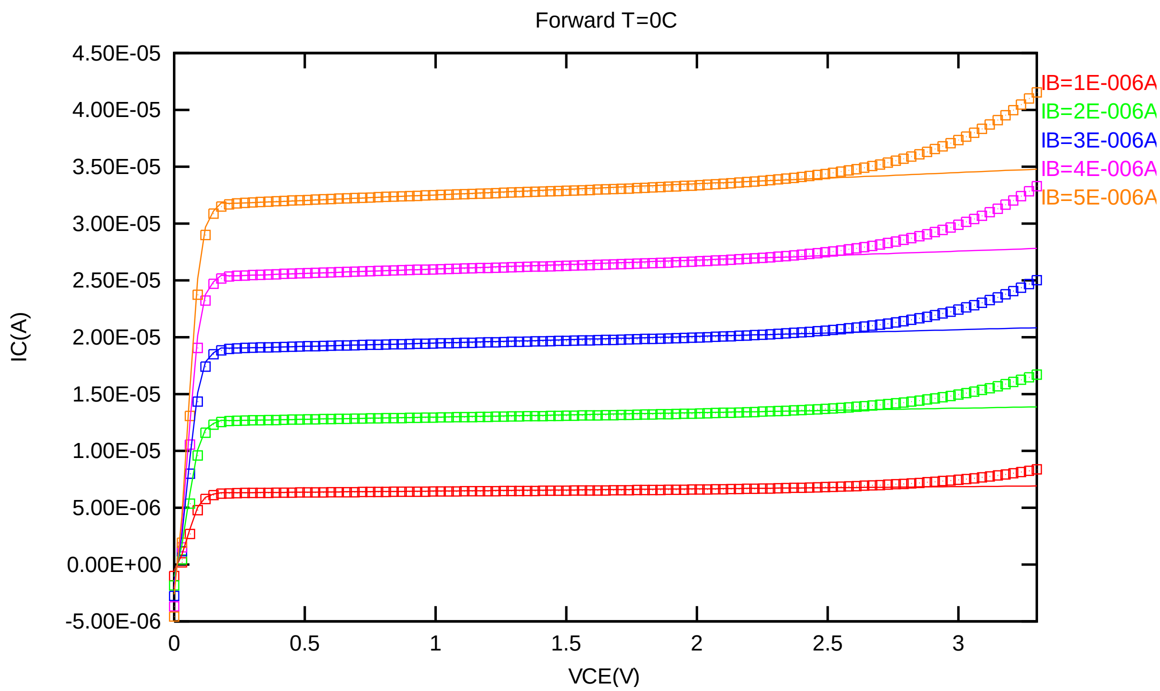


**Figure 2-32 Forward output characteristics of NPN_SV135X135_L110AE at 0 °C**

### 2.5.5 NPN_SV135X135_L110AE model verification at -40 °C

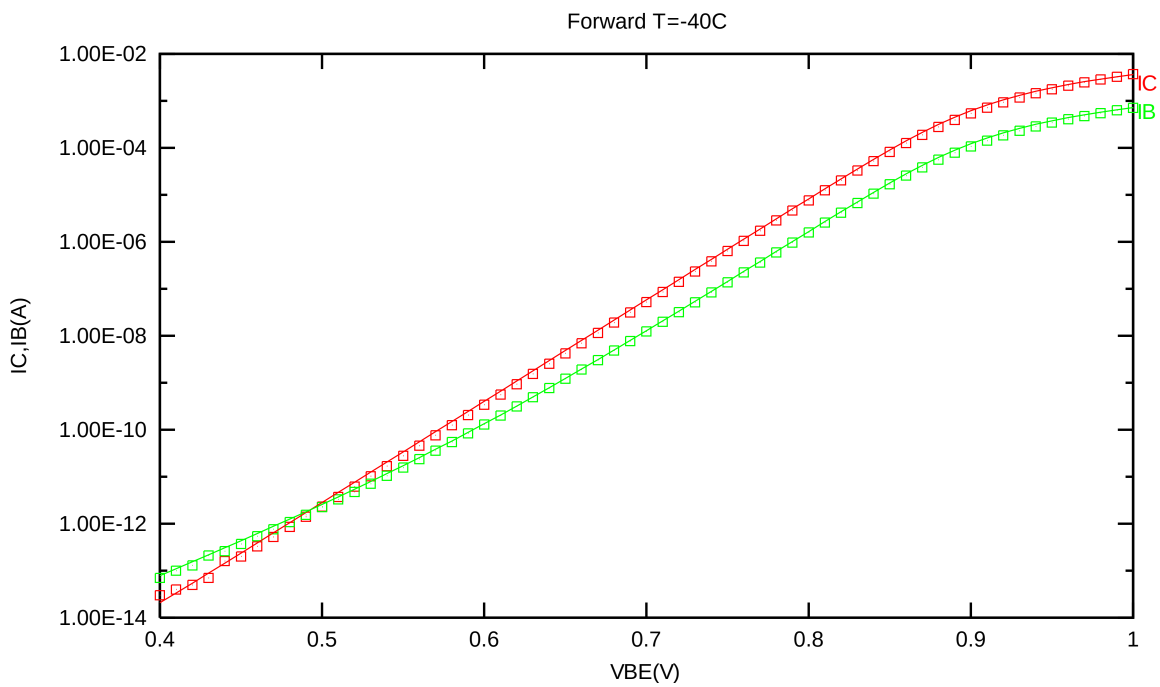


**Figure 2-33 Forward gummel plot of NPN_SV135X135_L110AE at -40 °C, VCB = 1V**

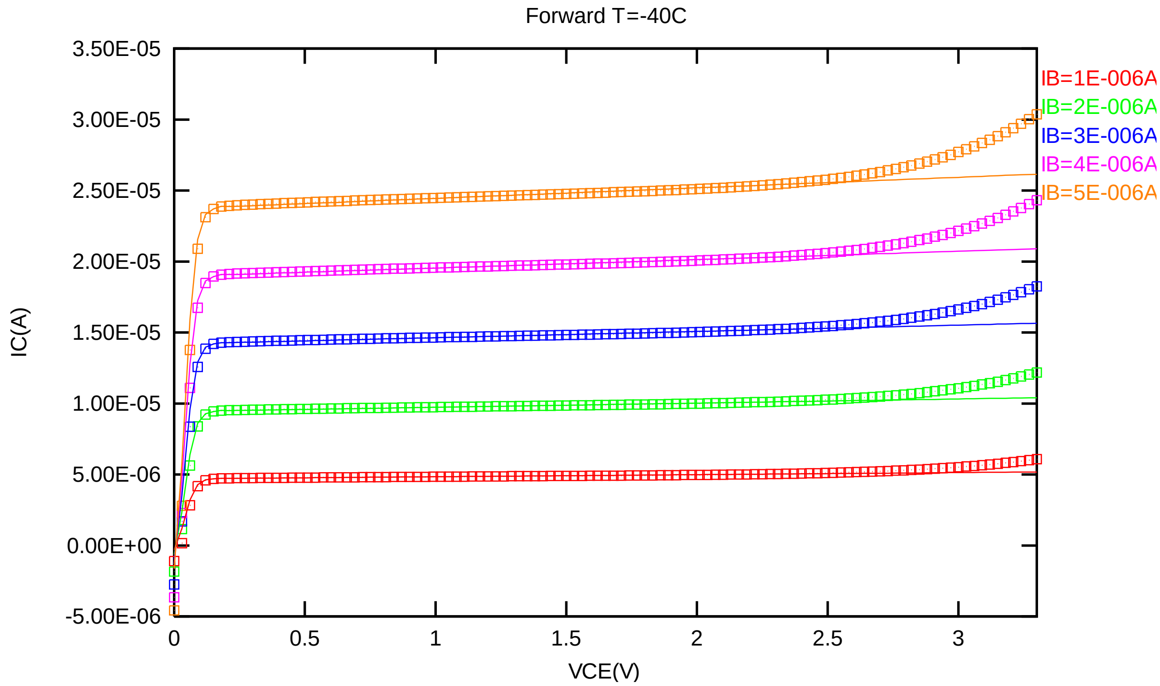


**Figure 2-34 Forward output characteristics of NPN_SV135X135_L110AE at -40 °C**

## 2.5.6 NPN_SV135X135_L110AE NPN model temperature effect

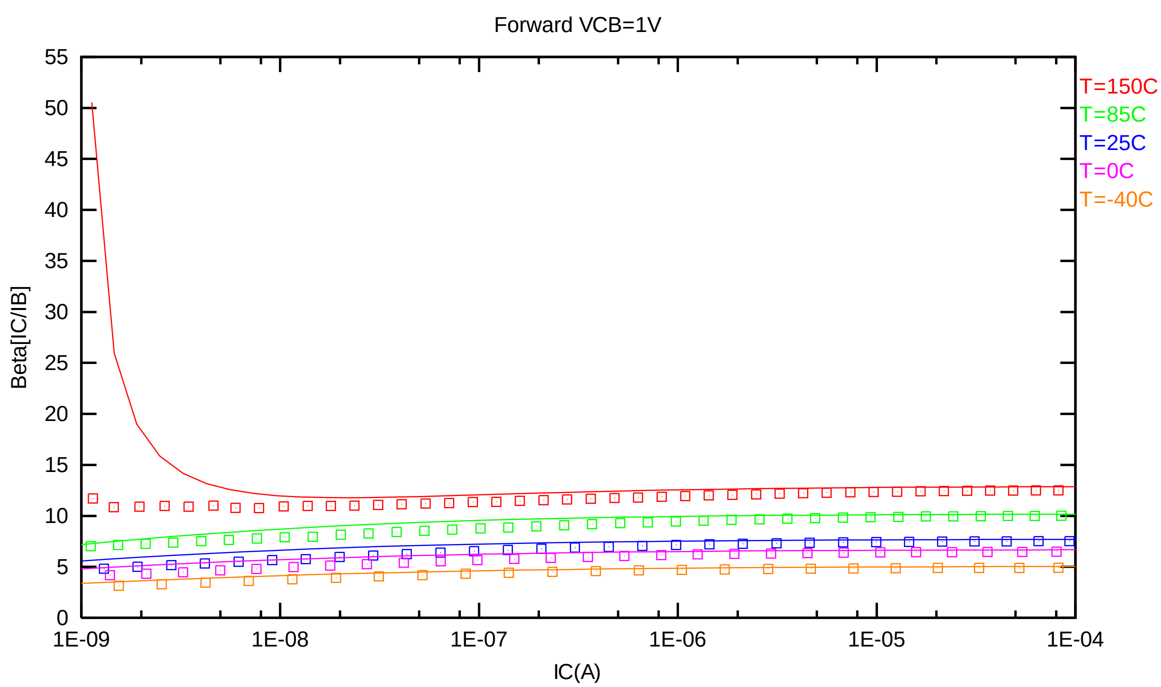


**Figure 2-35 Current gain vs. IC at different temperatures of NPN_SV135X135_L110AE, VCB = 1V**

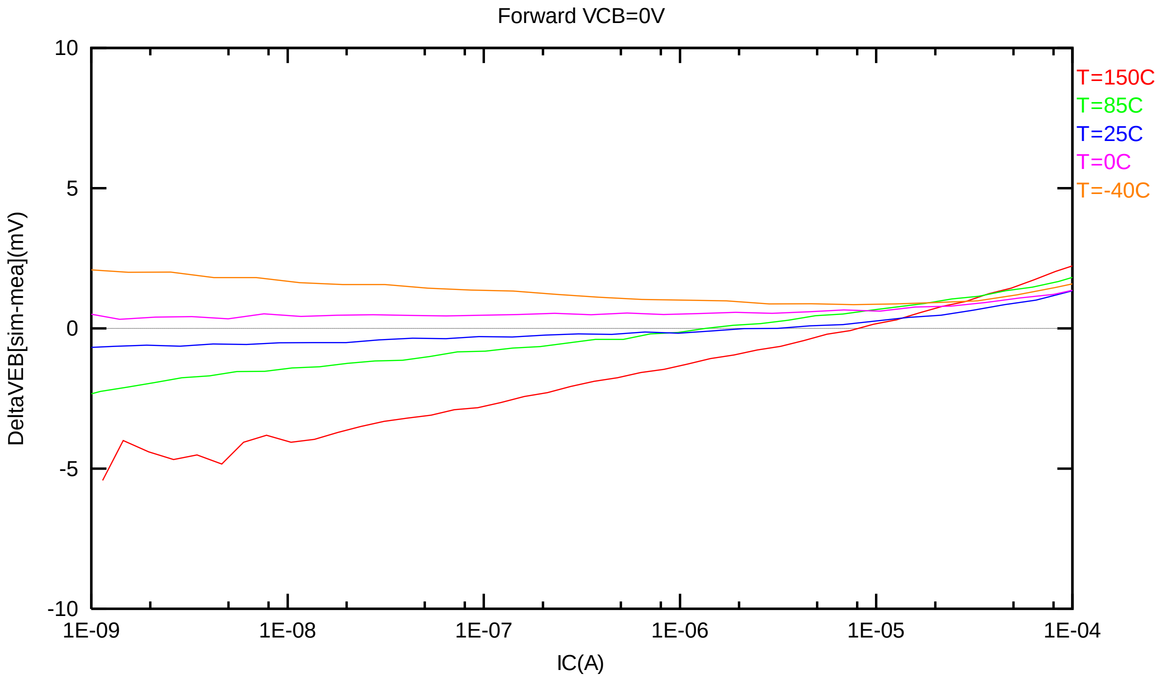


**Figure 2-36 DeltaVEB vs. IC at different temperatures of NPN_SV135X135_L110AE, VCB = 1V**

# 3 Worst Case Model

## 3.1 Parameter Variations

**Table 3-1 Corner parameters of NPN_SV45X45_L110AE**

| | Parameters | TT | FF | SS |
|---|---|---|---|---|
| 1 | cjc | 1.59710E-13 | x0.9000 | x1.1000 |
| 2 | cje | 1.51470E-18 | x0.9000 | x1.1000 |
| 3 | is | 3.8313E-18 | x0.8000 | x1.2000 |
| 4 | re | 8.0113E+00 | x0.8000 | x1.2000 |
| 5 | rc | 2.1391E+01 | x0.8000 | x1.2000 |
| 6 | bf | 8.8952E+00 | x1.2000 | x0.8000 |
| 7 | nf | 9.9778E-01 | x0.9950 | x1.0040 |
| 8 | ise | 1.7954E-16 | x0.7500 | x1.2500 |

**Table 3-2 Corner parameters of NPN_SV90X90_L110AE**

| | Parameters | TT | FF | SS |
|---|---|---|---|---|
| 1 | cjc | 3.06189E-13 | x0.9000 | x1.1000 |
| 2 | cje | 6.05880E-18 | x0.9000 | x1.1000 |
| 3 | is | 1.6257E-17 | x0.8000 | x1.3000 |
| 4 | re | 5.9003E+00 | x0.8000 | x1.2000 |
| 5 | rc | 2.0623E+01 | x0.8000 | x1.2000 |
| 6 | bf | 8.0543E+00 | x1.2000 | x0.8000 |
| 7 | nf | 1.0040E+00 | x0.9950 | x1.0050 |
| 8 | ise | 4.8428E-16 | x0.7500 | x1.2500 |

**Table 3-3 Corner parameters of NPN_SV135X135_L110AE**

| | Parameters | TT | FF | SS |
|---|---|---|---|---|
| 1 | cjc | 4.99919E-13 | x0.9000 | x1.1000 |
| 2 | cje | 1.36323E-17 | x0.9000 | x1.1000 |
| 3 | is | 3.1350E-17 | x0.7000 | x1.2000 |
| 4 | re | 4.7526E+00 | x0.8000 | x1.2000 |
| 5 | rc | 2.0585E+01 | x0.8000 | x1.2000 |
| 6 | bf | 7.6854E+00 | x1.2000 | x0.8000 |
| 7 | nf | 9.9800E-01 | x0.9950 | x1.0050 |
| 8 | ise | 7.8135E-16 | x0.7500 | x1.2500 |

# 4 Appendix

## 4.1 HSPICE Warning Message

**Table 4-1 HSPICE warning message**

| Item | Warning Message | Explanation |
|---|---|---|
| 1. | None | None |

## 4.2 ELDO Warning Message

**Table 4-2 ELDO warning message**

| Item | Warning Message | Explanation |
|---|---|---|
| 1. | None | None |

## 4.3 SPECTRE Warning Message

**Table 4-3 SPECTRE warning message**

| Item | Warning Message | Explanation |
|---|---|---|
| 1. | None | None |